



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO KAMALNATH FUEL STATION

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

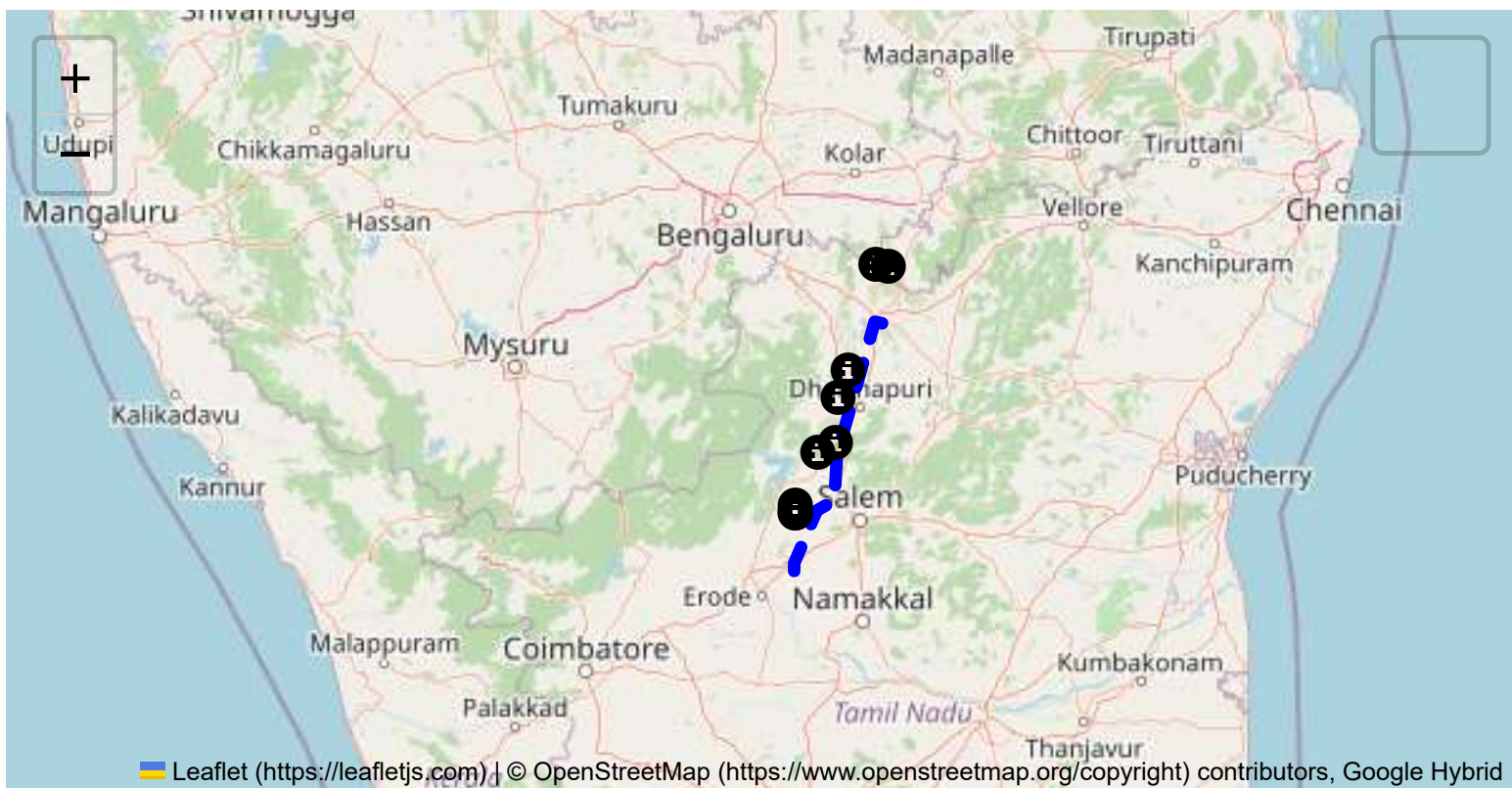
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

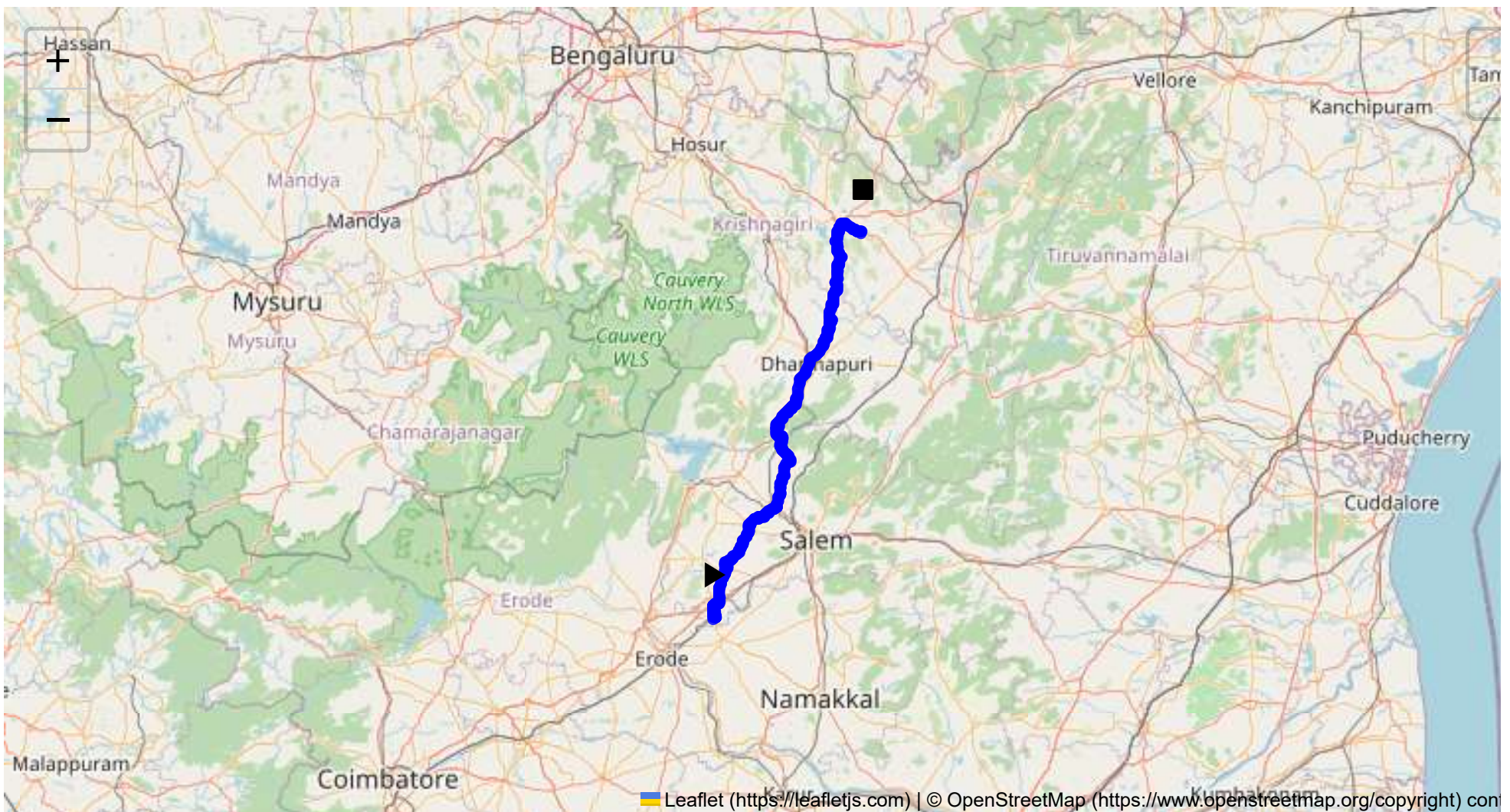
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 147.96 km**  
**Estimated Duration: 2.9 hours**  
**Adjusted Duration (Heavy Vehicle): 3.7 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.492564, 78.282379)**



## Welcome to the Journey Risk Management Study

Analyzing the proposed route from CVQF+23W, Sangagiri, Tamil Nadu to F7VJ+5V, Chinnapanamudlu, Tamil Nadu, here's a detailed safety assessment:

### 1. Overview of the Route Map

- The route spans roughly 148 kilometers through key towns and areas in Tamil Nadu, including Sangagiri, Kangeyampalayam, Salem, Tharamangalam, Omalur, Thoppur, Seshampatti, and



Krishnagiri. It involves traveling primarily via state highways, which are crucial for regional connectivity.

## **2. Typical Weather Conditions and Hazards**

- The region experiences tropical weather with distinct dry and wet seasons. Monsoons (June to September) can lead to heavy rainfall, causing waterlogging and reduced visibility, which are key concerns for heavy vehicle navigation.
- During summer months, high temperatures may lead to road surface damage.

## **3. Traffic Patterns & Congestion**

- Peak traffic hours generally occur between 8:00-10:00 AM and 5:00-7:00 PM. Salem and Krishnagiri, being major junctions, tend to be congested during these times.
- Congestion-prone areas include highway intersections, toll plazas, and city centers.

## **4. Road Quality and Infrastructure**

- Overall, state highways are well-maintained, but some segments may have potholes or uneven surfaces, particularly post-monsoon.
- Watch for roadworks or extensions near rapidly developing areas like Salem.

## **5. Alternative Routes for Emergencies**

- In case of severe congestion or incidents, the NH 44 via Salem can serve as a broader highway route, although it may be longer in terms of distance.
- Local bypasses around urban centers can provide temporary relief from congestion.

## **6. Local Regulations for Hazardous Material Transport**

- Transporting hazardous materials requires compliance with specific state and national regulations, including display of hazard placards and adherence to designated routes where heavy vehicle traffic is permitted.
- Night travel may be restricted in certain areas, so check local ordinances.

## **7. Historical Incidents Involving Heavy Vehicles**

- There have been reports of road accidents involving heavy vehicles in this region, typically due to driver fatigue, brake failures on slopes, or misjudgment in busy areas.
- The Salem to Dharmapuri stretch has witnessed incidents related to overspeeding.

## **8. Environmental Considerations and Sensitive Areas**

- Certain sections near Thoppur are ecologically sensitive and part of forest ranges. Be alert to wildlife crossings and ensure that hazardous material transport follows governmental guidelines.
- Avoid disposing of any waste materials along the route, particularly near water bodies.

## **9. Communication Coverage and Potential Dead Zones**

- The route generally has good communication coverage, but certain hilly areas near Dharmapuri might experience weak signals.
- Familiarize drivers with alternative communication methods like radio or satellite phones in these areas.

## **10. Estimated Emergency Response Times**

- Urban centers such as Salem and Krishnagiri have quicker response times (approximately 15-30 minutes), while more remote sections of the route could experience delays of up to an hour.
- Emergency services are generally well-organized in towns, with hospitals and firefighting units accessible within major towns.

11. Overall Summary of Risk Assessment

- This route is generally safe for transport of hazardous materials, assuming adherence to regulations and cautious driving, especially during adverse weather or peak traffic.
- Attention to potential roadworks, weather conditions, and congestion hotspots is crucial.
- Proactive route planning and regular communication can mitigate most risks, ensuring a smooth journey for truck drivers.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates           | Speed Limit |
|----|------------|------------|-----------------------|-------------|
| 0  | U-Turn     | High       | 11.4749755, 77.869149 | 10 KM/Hr    |
| 1  | Turn       | High       | 11.43968, 77.87345    | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44028, 77.87541    | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44898, 77.87410    | 15 KM/Hr    |
| 4  | Turn       | High       | 11.45007, 77.87201    | 15 KM/Hr    |
| 5  | Turn       | Medium     | 11.46392, 77.87328    | 30 KM/Hr    |
| 6  | Turn       | High       | 11.47391, 77.86844    | 15 KM/Hr    |
| 7  | Blind Spot | Blind Spot | 11.47481, 77.86920    | 10 KM/Hr    |
| 8  | Turn       | High       | 11.47464, 77.86938    | 15 KM/Hr    |
| 9  | Turn       | Medium     | 11.48005, 77.88450    | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.48385, 77.88782    | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.49155, 77.88776    | 30 KM/Hr    |
| 12 | Turn       | High       | 11.49469, 77.88488    | 15 KM/Hr    |
| 13 | Turn       | High       | 11.55614, 77.89764    | 15 KM/Hr    |
| 14 | Turn       | Medium     | 11.55680, 77.89969    | 30 KM/Hr    |
| 15 | Turn       | High       | 11.58218, 77.90845    | 15 KM/Hr    |
| 16 | Turn       | High       | 11.58330, 77.90642    | 15 KM/Hr    |
| 17 | Turn       | Medium     | 11.59189, 77.90645    | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.59328, 77.90809    | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.63867, 77.94841    | 30 KM/Hr    |
| 20 | Turn       | High       | 11.64217, 77.95200    | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 21 | Turn      | Medium     | 11.64712, 77.95118 | 30 KM/Hr    |
| 22 | Turn      | High       | 11.69597, 77.97126 | 15 KM/Hr    |
| 23 | Turn      | Medium     | 11.69794, 77.97520 | 30 KM/Hr    |
| 24 | Turn      | Medium     | 11.71380, 77.99532 | 30 KM/Hr    |
| 25 | Turn      | Medium     | 11.71731, 78.00006 | 30 KM/Hr    |
| 26 | Turn      | Medium     | 11.71947, 78.01172 | 30 KM/Hr    |
| 27 | Turn      | Medium     | 11.72472, 78.01630 | 30 KM/Hr    |
| 28 | Turn      | Medium     | 11.72760, 78.02285 | 30 KM/Hr    |
| 29 | Turn      | Medium     | 11.73761, 78.03437 | 30 KM/Hr    |
| 30 | Turn      | Medium     | 11.73759, 78.03658 | 30 KM/Hr    |
| 31 | Turn      | High       | 11.73652, 78.03863 | 15 KM/Hr    |
| 32 | Turn      | Medium     | 11.74252, 78.04705 | 30 KM/Hr    |
| 33 | Turn      | Medium     | 11.77968, 78.05777 | 30 KM/Hr    |
| 34 | Turn      | Medium     | 11.86887, 78.08377 | 30 KM/Hr    |
| 35 | Turn      | Medium     | 11.87976, 78.08182 | 30 KM/Hr    |
| 36 | Turn      | Medium     | 11.92621, 78.06514 | 30 KM/Hr    |
| 37 | Turn      | Medium     | 11.93719, 78.05372 | 30 KM/Hr    |
| 38 | Turn      | High       | 11.95828, 78.04983 | 15 KM/Hr    |
| 39 | Turn      | Medium     | 11.95997, 78.04946 | 30 KM/Hr    |
| 40 | Turn      | Medium     | 11.96335, 78.05141 | 30 KM/Hr    |
| 41 | Turn      | High       | 11.96408, 78.05131 | 15 KM/Hr    |
| 42 | Turn      | Medium     | 11.96511, 78.04966 | 30 KM/Hr    |
| 43 | Turn      | High       | 11.96608, 78.04919 | 15 KM/Hr    |
| 44 | Turn      | Medium     | 11.96819, 78.05088 | 30 KM/Hr    |
| 45 | Turn      | Medium     | 12.16993, 78.16780 | 30 KM/Hr    |
| 46 | Turn      | Medium     | 12.19885, 78.18158 | 30 KM/Hr    |
| 47 | Turn      | Medium     | 12.22467, 78.19138 | 30 KM/Hr    |
| 48 | Turn      | Medium     | 12.24611, 78.19519 | 30 KM/Hr    |
| 49 | Turn      | Medium     | 12.25061, 78.19978 | 30 KM/Hr    |
| 50 | Turn      | Medium     | 12.25349, 78.20037 | 30 KM/Hr    |
| 51 | Turn      | Medium     | 12.29134, 78.19920 | 30 KM/Hr    |
| 52 | Turn      | Medium     | 12.40597, 78.22280 | 30 KM/Hr    |
| 53 | Turn      | Medium     | 12.45026, 78.21827 | 30 KM/Hr    |
| 54 | Turn      | High       | 12.50709, 78.22230 | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 55 | Turn      | High       | 12.51556, 78.23049 | 15 KM/Hr    |
| 56 | Turn      | Medium     | 12.51540, 78.24044 | 30 KM/Hr    |
| 57 | Turn      | Medium     | 12.49340, 78.28179 | 30 KM/Hr    |
| 58 | Turn      | Medium     | 12.49293, 78.28194 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                              | coordinates            | speed_limit | risk_level |
|----|----------|-----------------------------------|------------------------|-------------|------------|
| 0  | hospital | Thilagam Hospital                 | 11.4753573, 77.8685829 | 30 km/h     | Medium     |
| 1  | clinic   | Meyyazhagan Clinic                | 11.4752254, 77.8685671 | 30 km/h     | Medium     |
| 2  | hospital | Dr. Jaganathan Hospital           | 11.4752717, 77.8709312 | 30 km/h     | Medium     |
| 3  | hospital | Kathir Eye Hospital               | 11.4740976, 77.8681591 | 30 km/h     | Medium     |
| 4  | police   | Police Quarters                   | 11.4741243, 77.8673495 | 30 km/h     | Medium     |
| 6  | hospital | Sankari Hospital                  | 11.4777496, 77.873892  | 30 km/h     | Medium     |
| 7  | clinic   | Dr. Dhanasekar Clinic             | 11.4793113, 77.8829247 | 30 km/h     | Medium     |
| 8  | hospital | Sri Vellaiyan Hospital            | 11.4798876, 77.8831394 | 30 km/h     | Medium     |
| 9  | clinic   | Sun Eye Care                      | 11.6936242, 77.9730948 | 30 km/h     | Medium     |
| 10 | hospital | Dr. Pandiyarajan Hospital         | 11.692704, 77.969344   | 30 km/h     | Medium     |
| 11 | hospital | Dr. Sengodan Hospital             | 11.6959527, 77.9695325 | 30 km/h     | Medium     |
| 12 | hospital | Omalur Government Hospital        | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 13 | clinic   | R.R. Nursing Home                 | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 14 | hospital | Sri Hari Hospital                 | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 15 | hospital | Sugumar Hospital                  | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 16 | hospital | Sri Sugam Hospital                | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 17 | hospital | Saravana Hospital                 | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 18 | hospital | Arun Hospital                     | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 19 | clinic   | Saratha Clinic                    | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 21 | clinic   | Sri Meenakshi Nursing Home, Salem | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 22 | clinic   | Divyam Child Care Clinic          | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 23 | clinic   | Momota Clinic                     | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 24 | clinic   | Gomathi Clinic                    | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 25 | hospital | Vinayaka Child Hospital           | 11.745812, 78.0480225  | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 26 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 27 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 28 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 29 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 30 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 31 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 32 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 33 | hospital | St. Joseph's Speciality Hospital   | 11.761611, 78.0522344  | 30 km/h     | Medium     |
| 35 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 37 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 38 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 39 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |
| 41 | hospital | Government Hospital                | 12.3048002, 78.2060881 | 30 km/h     | Medium     |
| 46 | hospital | TCR Multi Speciality Hospital      | 12.5088183, 78.2237657 | 30 km/h     | Medium     |

### Crowded Spots

|    | type    | name                                              | coordinates            | speed_limit | risk_level |
|----|---------|---------------------------------------------------|------------------------|-------------|------------|
| 5  | school  | Goverment Boys Higher Secondary School            | 11.477659, 77.8659632  | 30 km/h     | Medium     |
| 20 | school  | Fathima Hr Sec School                             | 11.7410508, 78.0479272 | 30 km/h     | Medium     |
| 34 | school  | John Britto Matriculation Higher Secondary School | 11.7676907, 78.0513092 | 30 km/h     | Medium     |
| 36 | college | Dhirajlal Gandhi College of Technology            | 11.7968014, 78.0562095 | 30 km/h     | Medium     |
| 40 | college | Laxminarayana Polytechnic College                 | 12.0421277, 78.103203  | 30 km/h     | Medium     |
| 42 | school  | Government Boys Higher secondary school           | 12.3047059, 78.2065495 | 30 km/h     | Medium     |
| 43 | school  | Nalanda Metric Higher Secondary School            | 12.488407, 78.21856    | 30 km/h     | Medium     |
| 44 | school  | Nalanda International Public School               | 12.4994979, 78.2235094 | 30 km/h     | Medium     |



|    | type   | name                    | coordinates              | speed_limit | risk_level |
|----|--------|-------------------------|--------------------------|-------------|------------|
| 45 | school | Don Bosco School Ground | 12.505954,<br>78.2222003 | 30 km/h     | Medium     |

## Route Photos of Risky Spots

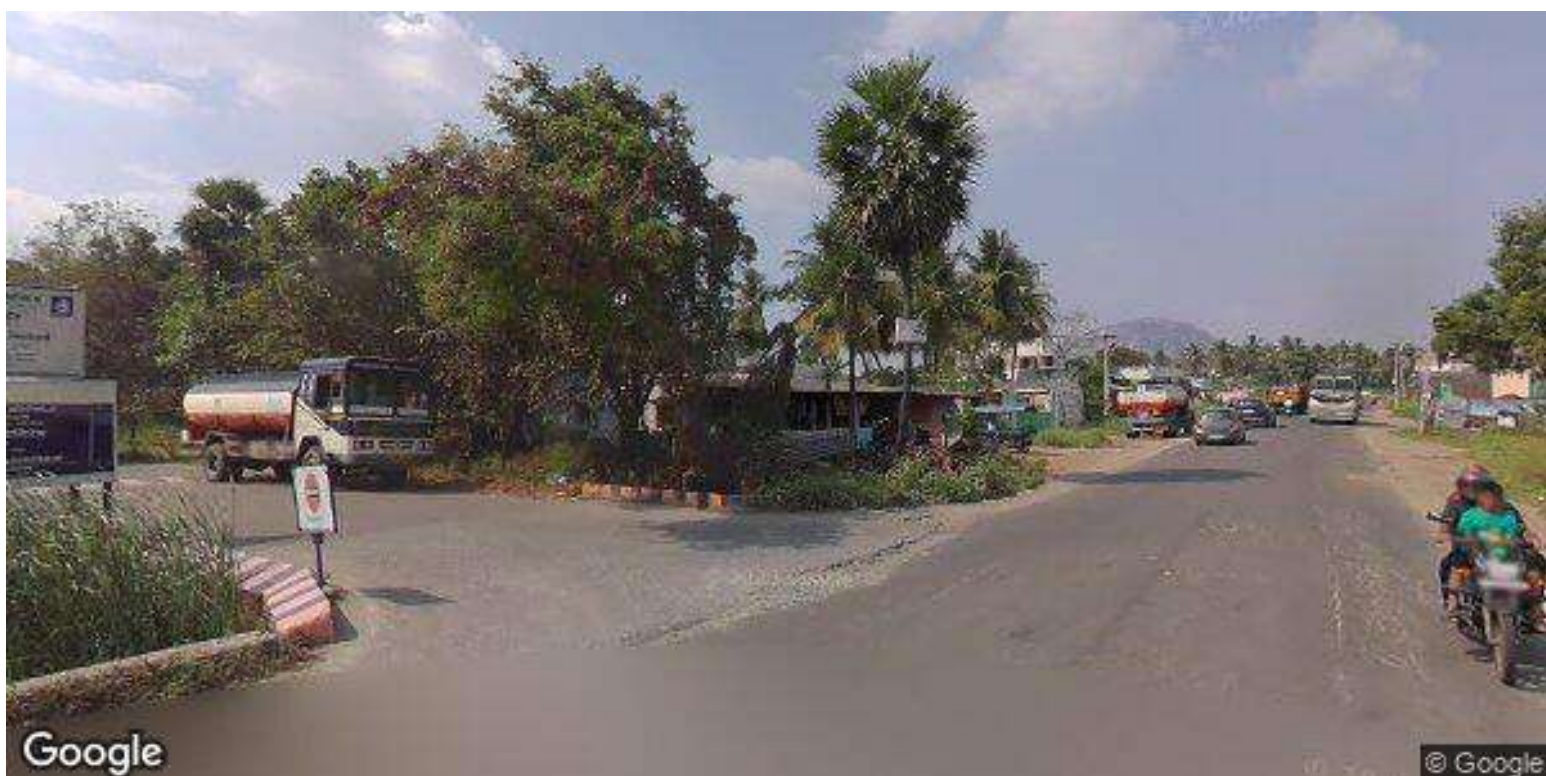


**Risk Type:** U-Turn

**Risk Level:** High

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.4749755, 77.869149



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44028, 77.87541





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44898, 77.87410



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45007, 77.87201





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.46392, 77.87328**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.47391, 77.86844**



**Risk Type: Blind Spot**

**Risk Level: Blind Spot**

**Speed Limit: 10 KM/Hr**

**Coordinates: 11.47481, 77.86920**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.47464, 77.86938



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.48005, 77.88450





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.48385, 77.88782



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49155, 77.88776



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.49469, 77.88488





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.55614, 77.89764



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.58330, 77.90642





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.59189, 77.90645



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.59328, 77.90809



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.63867, 77.94841





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.64217, 77.95200



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64712, 77.95118





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.69597, 77.97126



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.69794, 77.97520



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71380, 77.99532





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71731, 78.00006



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71947, 78.01172





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.72472, 78.01630



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.72760, 78.02285



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73761, 78.03437





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73759, 78.03658



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.73652, 78.03863





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.74252, 78.04705**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.86887, 78.08377**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.87976, 78.08182**





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92621, 78.06514



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93719, 78.05372





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95828, 78.04983



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95997, 78.04946



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96335, 78.05141





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96408, 78.05131



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96511, 78.04966





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96608, 78.04919



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96819, 78.05088



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.16993, 78.16780





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.19885, 78.18158



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.22467, 78.19138





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.24611, 78.19519



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25061, 78.19978



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25349, 78.20037





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.29134, 78.19920



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.40597, 78.22280





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 12.45026, 78.21827**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 12.50709, 78.22230**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 12.51540, 78.24044**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.49340, 78.28179



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.49293, 78.28194

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